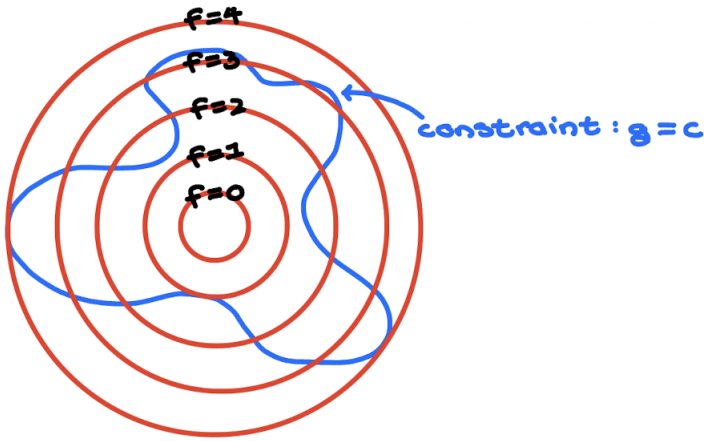


A. **Lagrange Multipliers.** We aim to optimize $f(\vec{x})$ subject to **equality** constraint

Consider for example the contours of f depicted along with the constraint.



An extremizer of f subject to $g = c$ **cannot** occur at a point where

as motion along $g = c$ in one direction will:

an in the other direction will:

Thus an extremizer of f subject to $g = c$ **must** occur at a point P where

Because gradients point:

Another way of phrasing this is

This alternative characterization requires:

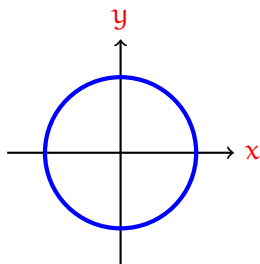
The scenario $\vec{\nabla}g = \vec{0}$ on the constraint set is not typical, and we call it:

The method of **Lagrange multipliers** searches for maximizers/minimizers by

and will succeed if there is a nondegenerate maximizer/minimizer.

Example 1. Find the maximum/ minimum of $f(x, y) = xy$ subject to $x^2 + y^2 = 2$.

Assume an absolute minimizer/minimizer exists and is nondegenerate.



Solution. The assumption implies Lagrange multipliers will succeed.

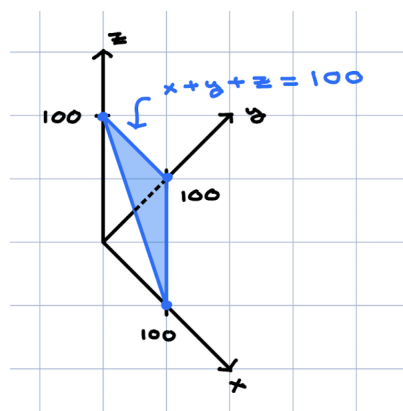
The constraint function is:

We set up and solve system

We evaluate f at the solutions.

Example 2. Find the closest point on the plane $x + y + z = 100$ to the origin.

Assume a closest point exists and is nondegenerate for your optimizing function.



Solution. The assumption implies Lagrange multipliers will succeed.

The constraint function is:

We wish to minimize distance from (x, y, z) to the origin, given by

The square-root makes differentiation tedious.

To avoid it, let's locate the closest point by instead minimizing:

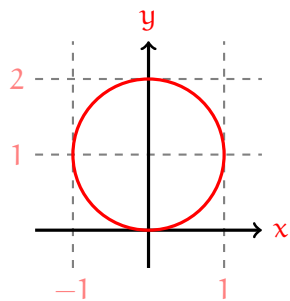
$D =$

We set up and solve system

We evaluate d at the solution.

Example 3. Maximize/minimize $f(x, y) = 2x^2 + \frac{y^3}{3} - y$ subject to $x^2 + (y-1)^2 = 1$.

You may assume a nondegenerate maximizer/minimizer exists.



Solution. The assumption implies Lagrange multipliers will succeed.

The constraint function is:

We set up and solve system

We evaluate f at the solutions.